

(a) Fig. 9.1 is a diagram of a relaxed sarcomere in striated muscle.

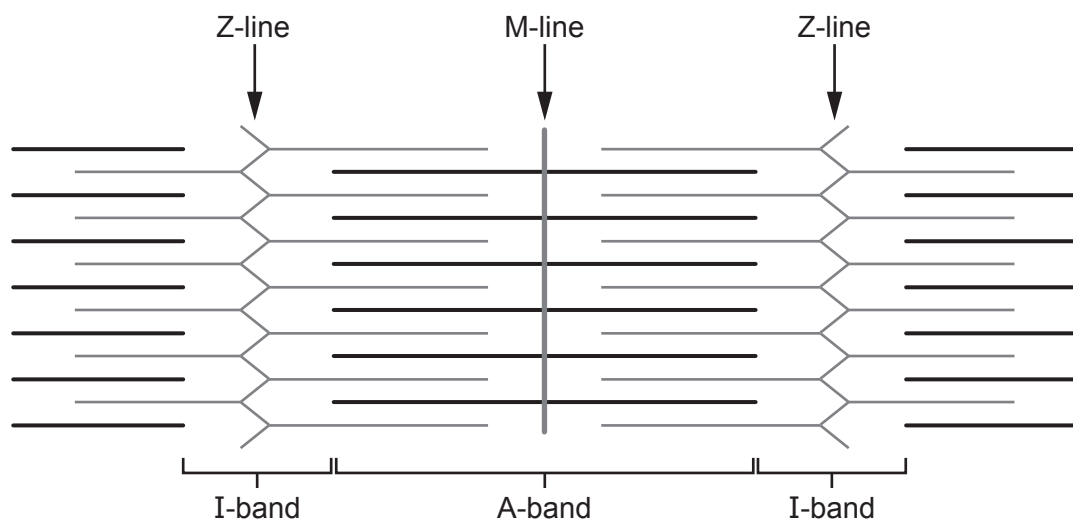


Fig. 9.1

- (i) On Fig. 9.1, use label lines and letters to label:
- an actin filament with the letter **P**
 - a myosin filament with the letter **R**.

[2]

- (ii) State what happens to the A-band and the I-band when the sarcomere contracts.

A-band

I-band

[2]

- (b)** The plant *Strychnos toxifera* produces the toxin curare, which can cause muscle paralysis in mammals.

The toxin acts by binding to receptors on the cell surface membranes (sarcolemma) of muscle cells at neuromuscular junctions.

- (i) Suggest how binding of curare to receptors may cause muscle paralysis.

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[4]

(ii) Suggest why the action of curare may lead to the death of a mammal.

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..... [2]

[Total: 10]